Formatting with LaTeX

IEEE EMBS, 2026

Why would you even use LaTeX when Word and Docs are right there?

Short Answer: trust me

Long Answer: LaTeX can improve your text formatting for reports, projects, or even personal documents like resumes

datacamp

LaTeX Cheat Sheet

Learn data & AI skills online at www.DataCamp.com

> Comments

Insert a comment with %

```
% this text is ignored
```

> Environments

Environments are sections in a document that apply special behavior rules to the enclosed text.

```
\begin{environment-name}
% environment contents
\end{environment-name}
```

> Equations

Description	Code	Output
Insert inline equation with $	`The equation is $e=mc^2$`	The equation is $e=mc^2$
Insert inline equation with \(	`The equation is \(e=mc^2\).`	The equation is $e=mc^2$.
Insert unnumbered equation with $$	`The equation is $$ e=mc^2 $$`	The equation is $$e=mc^2$$
Insert unnumbered equation with \[	`The equation is \[ e=mc^2 \]`	The equation is $$e=mc^2$$
Insert numbered equation with \begin{equation}	`\begin{equation} e=mc^2 \end{equation}`	$e=mc^2$ (1)

> Document structure

Preamble

Preamble content must appear before `\begin{document}`.
Define the document class with `\documentclass`.

```
\documentclass[options]{class}
```

Common class values

Class	Description
article	Default class for composing an article.
report	Typeset a multi-chapter report.
book	A class for typesetting books.
beamer	For producing presentations and slides.

Common class options

Option	Default value	Available values
fontsize	11pt	10pt, 12.5pt, etc
paper	a4, portrait	a5, letter, landscape
toc	nolistof	listof, listofnumbered
captions	tablebelow, figurebelow	tableabove, figureabove
twocolumn	false	true

Loading Packages

Load a package with `\usepackage`

```
\usepackage[options]{package}
```

Common package examples

Functionality	Code
Mathematical typesetting	`\usepackage{amsmath}`
Set the text encoding	`\usepackage[utf8]{inputenc}`
Multilingual support	`\usepackage[english, spanish, portuguese]{babel}`

Document

Include the entire document in a document environment.

```
\begin{document}
% The complete document contents
\end{document}
```

Title Page

Element	Output
Title of document	`\title{text}`
Author name	`\author{text}`
Date	`\date{\today}`
Generate the title page	`\maketitle`

Content Listings

Element	Output
Table of contents	`\tableofcontents`
List of tables	`\listoftables`
List of figures	`\listoffigures`

Headings

Element	Code
Chapter	`\chapter{The Last Chapter}`
Section	`\section{Conclusion}`
Subsection	`\subsection{Final thoughts}`

Text Justification

Description	Code	Output
Left align	`\begin{flushleft}`	On the left
Center align	`\begin{center}`	In the middle
Right align	`\begin{flushright}`	On the right

Lists

Description	Code	Output
Bulleted list	`\begin{itemize} \item flour \item milk \item eggs \end{itemize}`	• flour • milk • eggs
Numbered list	`\begin{enumerate} \item Learn LaTeX \item ??? \item Profit \end{enumerate}`	1. Learn LaTeX 2. ??? 3. Profit
Named elements in list	`\begin{description} \item[datum] Single piece of information \item[data] Plural of datum \end{description}`	**datum** Single piece of information **data** Plural of datum

> Text Mode

Formatting

Description	Code	Output
Bold	`\textbf{To boldly go}`	To **boldly go**
Italic	`\textit{Leaning Tower of Pisa}`	*Leaning Tower of Pisa*
Small caps	`\textsc{caveman}`	CAVEMAN

Spaces

Description	Code	Output
Thinnest space	`very\,close`	very close
Thin space	`quite\:close`	quite close
Medium space	`fairly\;close`	fairly close
1 em space	`quite\quad far`	quite far
2 em space	`very\qquad far`	very far

Accents

Description	Code	Output
Hat	`\hat{a}`	â
Bar	`\bar{a}`	ā
Tilde	`\tilde{o}`	õ

> MathMode

Mathematical Arithmetic Operators

Description	Code	Output
Plus	`2 + 3`	$2 + 3$
Minus	`9 - 6`	$9 - 6$
Plus or Minus	`\pm 1.55`	± 1.55
Multiplication (times)	`6 \times 7`	6×7
Multiplication (dot)	`6 \dot 7`	$6 \dot 7$
Division (slash)	`7 / 22`	$7 / 22$
Division (symbol)	`7 \div 22`	$7 \div 22$

Numeric Comparison Operators

Description	Code	Output
Equal	`1 + 2 = 3`	$1+2=3$
Not equal	`1 + 2 \neq 4`	$1+2 \neq 4$
Approximately equal	`\pi \approx 3.14`	$\pi \approx 3.14$
Less than	`6 < 7`	$6 < 7$
Less than or equal to	`6 \leq 7`	$6 \leq 7$
Greater than	`8 > 7`	$8 > 7$
Greater than or equal to	`8 \geq 7`	$8 \geq 7$

Equations

Description	Code	Output
Fraction	`\frac{7}{22}`	$\frac{7}{22}$
Square root	`\sqrt{49}`	$\sqrt{49}$
Nth root	`\sqrt[5]{243}`	$\sqrt[5]{243}$
Subscript	`x_{i}`	x_i
Superscript	`x^{3}`	x^3
Absolute values	`|x|`	$\|x\|$
Natural logarithm	`\ln(5)`	$\ln(5)$
Logarithm	`\log_{2}8`	$\log_2 8$
Exponential	`\exp{x}`	e^x
Sine, cosecant	`\sin{x}=\frac{1}{\csc{x}}`	$\sin x = \frac{1}{\csc x}$
Cosine, secant	`\cos{x}=\frac{1}{\sec{x}}`	$\cos x = \frac{1}{\sec x}$
Tangent, cotangent	`\tan{x}=\frac{1}{\cot{x}}`	$\tan x = \frac{1}{\cot x}$

Greek and Hebrew Letters

Description	Code	Output
Lowercase delta	`\delta`	δ
Uppercase delta	`\Delta`	Δ
Aleph	`\Aleph`	$\aleph$

Set Theory

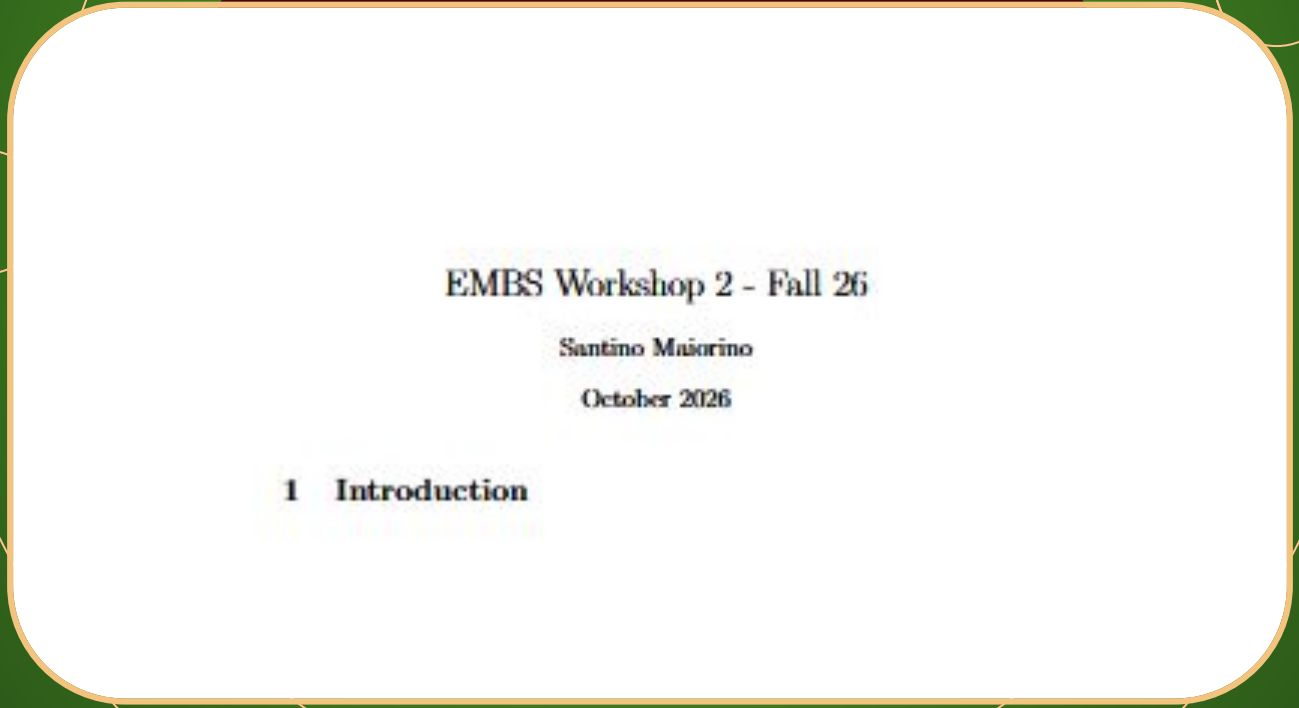
Description	Code	Output
Union	`\cup`	$\cup$
Intersection	`\cap`	$\cap$
Is a member of	`\in`	$\in$
Empty set	`\emptyset`	$\emptyset$

Logic Operators

Description	Code	Output
Not	`\neg A`	$\neg A$
And	`A \land B`	$A \land B$
Or	`A \lor B`	$A \lor B$

Learn more about LaTeX at www.DataCamp.com

Activity 0:
Make Your First Document in Overleaf!



EMBS Workshop 2 - Fall 26

Santino Maiorino

October 2026

1 Introduction

Answer to Activity 0

```
\documentclass{article}
\usepackage{graphicx} % Required for inserting images

\title{EMBS Workshop 2 - Fall 26}
\author{Santino Maiorino}
\date{October 2026}

\begin{document}

\maketitle

\section{Introduction}

\end{document}
```

Walkthrough: Title Page

```
\begin{document}

\title{Vickers Micro-Hardness Testing}
\author[1]{Santino Maiorino}
\date{\parbox{\linewidth}{\centering February 23, 2026 \endgraf
Contact: sa561579@ucf.edu \endgraf Group 2 (Section 0013)}} %parbox is
what you needed
% Affiliations (if needed)
\affil[1]{Department of Materials Science and Engineering, University
of Central Florida, Orlando, Florida, U.S.A.}

\maketitle
\newpage
```



Vickers Micro-Hardness Testing

Santino Maiorino[1]

[1]Department of Materials Science and Engineering, University of Central Florida, Orlando, Florida, U.S.A.

February 23, 2026
Contact: sa561579@ucf.edu
Group 2 (Section 0013)

Activity 1:
Headings and Subheadings

1 Introduction

1.1 Literature Review

Will insert sample citation here later.

1.2 Objective

2 Experiments

2.1 Instrumentation

2.2 Procedure

2.2.1 Technique A

2.2.2 Technique B

3 Results and Discussion

4 Conclusion

5 Acknowledgements

Answer to Activity 1

```
\newpage

% Section 1: Introduction
%
\section{Introduction}
\subsection{Literature Review}\label{Section-LitReview}
Will insert sample citation here later.

\subsection{Objective}

% Section 2: Experiments
%
\section{Experiments}\label{Section-Experiments}
\subsection{Instrumentation}
\subsection{Procedure}
\subsubsection{Technique A}
\subsubsection{Technique B}

% Section 3: Results and Discussion
%
\section{Results and Discussion}
```

```
% Section 4: Conclusion
%
\section{Conclusion}

% Section 5: Acknowledgements
%
\section{Acknowledgements}




\end{document}
```

Activity 2:
Adding Images

2 Experiments

2.1 Instrumentation



Figure 1: This is NOT an Anton Parr MCR 703 rheometer.

Hints: 1) create an images folder, 2) keep under 95% size, and 3) use [hbt!] to place your image under the Instrumentation subheading

Answer to Activity 2

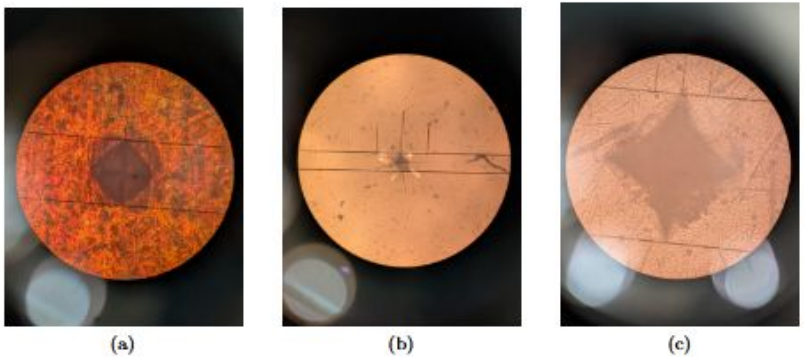
```
\section{Experiments}\label{Section-Experiments}
\subsection{Instrumentation}
\begin{figure}[hbt!] %
    \centering
    \includegraphics[width=0.75\linewidth]{images/TOTTMNT-Genghis-Frog-
    Error.jpg}
    \caption{This is NOT an Anton Parr MCR 703 rheometer.}
    \label{fig:Genghis-Freaking-Frog}
\end{figure}
```

Walkthrough: Multiple Images

```
\begin{figure}[hbt!]
    \centering

    \subfloat[\label{fig:Microscope-Cu}]
    {\includegraphics[width=0.3\textwidth]{Figs/Samples/
    Scope_Copper.jpg}}\hfill
    \subfloat[\label{fig:Microscope-Glass}]
    {\includegraphics[width=0.3\textwidth]{Figs/Samples/
    Scope_Glass.jpg}}\hfill
    \subfloat[\label{fig:Microscope-PVC}]
    {\includegraphics[width=0.3\textwidth]{Figs/Samples/
    Scope_PVC Yellow.jpg}}

    %General caption
    \caption{Microscope images of the Vickers hardness test
    indentations in the copper pipe (a), the glass slide (b),
    and the yellow PVC pipe (c). (Photos are courtesy of Andrew
    Riiska, with some edits made by the author for formatting.)}
    \label{fig:OtherSamples-Microscopy}
\end{figure}
```



(a) (b) (c)

Figure 7: Microscope images of the Vickers hardness test indentations in the copper pipe (a), the glass slide (b), and the yellow PVC pipe (c). (Photos are courtesy of Andrew Riiska, with some edits made by the author for formatting.)

Activity 3:
Formatting Tables

3 Results and Discussion

Table 1: Your First Table :3

1.9	3.2	6.2
4.7	2.4	9.3
6.3	2.3	3.3

This smaller caption size is a bit more tricky, so I can help you out

Hints: 1) still use [hbt!], 2) use the default “caption” command for your title, and 3) the commands \begin{tablenotes} and \footnotesize will come in handy for the caption

Answer to Activity 3

```
\begin{table}[hbt!]
    \centering
    \caption{Your First Table :3}
    \begin{tabular}{ccc}
        1.9 & 3.2 & 6.2\\
        4.7 & 2.4 & 9.3\\
        6.3 & 2.3 & 3.3\\
    \end{tabular}

    \begin{tablenotes}[para, flushcenter] % Caption here
    % If same table numbered twice, remove "\caption" from under table
    \centering
    \footnotesize
    This smaller caption size is a bit more tricky, so I can help you out
    \end{tablenotes}

\label{tab:BasicTable}
\end{table}
```

Walkthrough: More Sophisticated Table

Table 4: Vickers Measurements for Other Materials

Material	Load (kgf (lbs))	d_1 (μm)	d_2 (μm)	$HV_{observed}$ [1]	$HV_{expected}$ [1]	Error (%)
Carbon Fiber, sheet	3.0 (6.62)	336.0	380.7	43.3	43.3	0.049
Glass cover slide, sheet	3.0 (6.62)	108.6	108.7	471.3	471.3	0.009
Carbon, rod	3.0 (6.62)	356.8	354.1	44.0	44.0	0.071
Copper, pipe	10.0 (22.05)	379.1	375.0	130.4	130.4	0.028
PVC (white), pipe	5.0 (11.03)	772.9	821.0	14.6	14.6	0.011
PVC (yellow), pipe	5.0 (11.03)	737.0	780.4	16.1	16.1	0.046

The measurements and HV values for the samples not made of aluminum and steel.
[1] HV units: kg/mm^2.

Walkthrough: More Sophisticated Table

```
\begin{table}[hbt!]
    \centering
    \caption{Vickers Measurements for Other Materials} %title of table
    \hspace*{-2.1cm}\begin{tabular}{@{}ccccccc@{}}

    \toprule % draws line
    Material & Load (kgf (lbs)) & $d_1$ $(\mu \text{m})$  & $d_2$ $(\mu \text{m})$ & $\text{HV}_{observed}$ \footnotemark[1] & $\text{HV}_{expected}$ \footnotemark[1] & Error (\%)\\

    \midrule
    Carbon Fiber, sheet & 3.0 (6.62) & 336.0 & 380.7 & 43.3 & 43.3 & 0.049\\
    Glass cover slide, sheet & 3.0 (6.62) & 108.6 & 108.7 & 471.3 & 471.3 & 0.009\\
    Carbon, rod & 3.0 (6.62) & 356.8 & 354.1 & 44.0 & 44.0 & 0.071\\
    Copper, pipe & 10.0 (22.05) & 379.1 & 375.0 & 130.4 & 130.4 & 0.028\\
    PVC (white), pipe & 5.0 (11.03) & 772.9 & 821.0 & 14.6 & 14.6 & 0.011\\
    PVC (yellow), pipe & 5.0 (11.03) & 737.0 & 780.4 & 16.1 & 16.1 & 0.046\\

    \bottomrule
    \end{tabular}\hspace*{-2.1cm}

    \begin{tablenotes}[para, flushcenter] % Caption here
    \centering
    \footnotesize
    The measurements and HV values for the samples not made of aluminum and steel.\\
    $^1$ HV units: $\text{kg} / \text{mm}^2$.
    \end{tablenotes}

\label{tab:OtherSamples} %for referring to table
\end{table}
```

Activity 4:
Getting Used to Templates

Look up “Jake’s Resume Template” and “Springer Nature Latex Template”

Thank you!

Encourage your audience to provide feedback and continue the conversation at a later date. *(No thanks, I'm good! I can do that myself.)*

Contact
name@example.com

State the purpose of this presentation or meeting.
Explain why the content is relevant to your audience.
For example, your objective might be to better understand the competitive landscape to recognize business opportunities.

Meet the team



Full Name
Title

Write a brief bio of this team member. Mention their background, area of expertise, and how they contributed to the competitive analysis.



Full Name
Title

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Meet the team



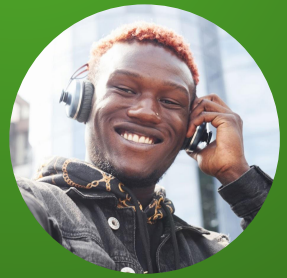
Full Name
Title



Full Name
Title



Full Name
Title



Full Name
Title



Full Name
Title



Full Name
Title

How we conducted our analysis

If you repeated steps or iterated on the process, change the conclusion step to show what you repeated.



Walk your audience through your process.

At each step, briefly explain the actions carried out by the team.

For example, they might have identified competitors, or defined analysis criteria.

Describe how they interviewed stakeholders, tested with users, and collected industry data.

Eventually, the team arrived at the analysis that you'll be presenting in the slides ahead.

How we conducted our analysis

1. Identified competitors

Split your research methodology into phases. For example, as your first step, you might have identified competitors. Walk your audience through this process. What criteria did you use? What kind of competitors did you focus on?

2. Gathered data

After identifying your competitors, the next step might have been gathering data. If so, what sources did you consult? Did you collect industry reports and customer reviews? Did you conduct surveys and interviews?

3. Analyzed and interpreted data

Finally, what did you do with all this information? How did you analyze and interpret it? Preview the results of your competitive analysis, which will be presented on the following slides.

Competitor landscape

Identify and classify your competitors. Place them into quadrants based on how successful they are now — and how prepared they are for the future.

Challengers	Leaders
List competitors who currently dominate the market, but are not prepared for the future.	List competitors who both dominate the current market and are well-positioned for the future.
Niche players	**Visionaries**
List competitors who are successful within their niche, but have no clear plan or roadmap.	List competitors who have a clear vision of the future, but have yet to fulfill their business potential.

If you add a diagram to this slide, explain what it shows. Help your audience make sense of the information.



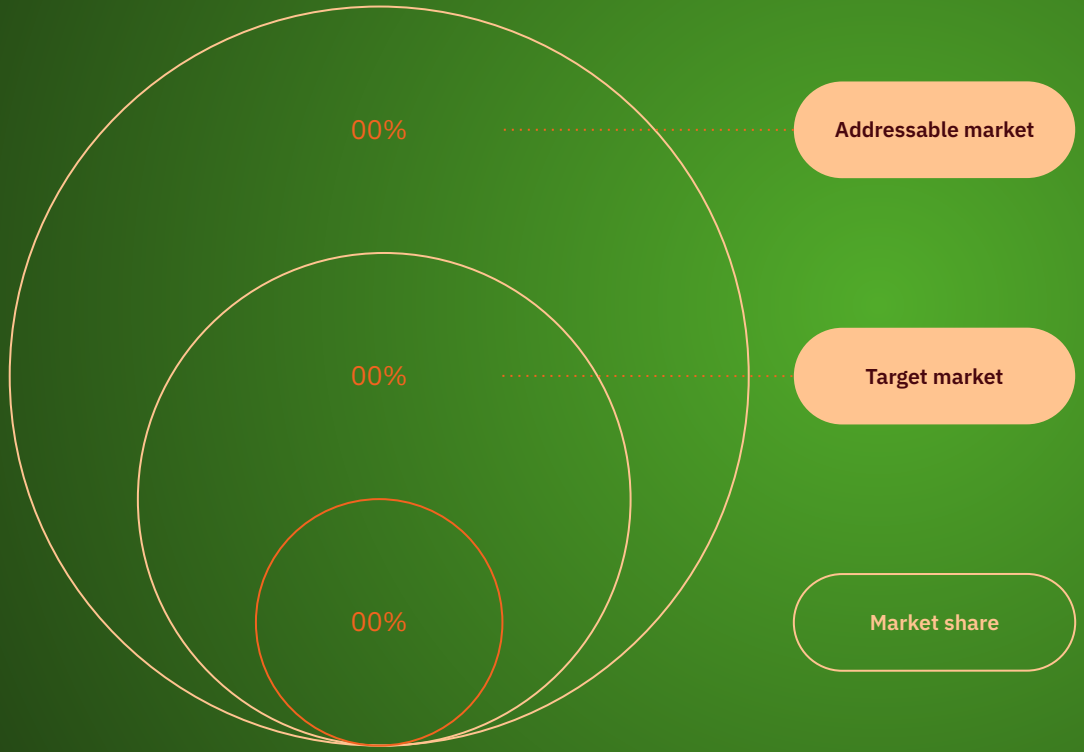
Competitor landscape

Map your competitors based on price and quality, using a brand positioning graph

If you add a diagram to this slide, explain what it shows. Help your audience make sense of what they're seeing.

Market size

Your market share, in context



If you add a diagram to this slide, explain what it shows. Help your audience make sense of what they’re seeing.

Market trends

Share your vision about where the market is heading, based on the latest trends

01

Identify a market trend

Explain why you consider this to be a current or future trend. Support your case with sales data, user interviews, media coverage, industry reports, etc.

02

Identify a market trend

Explain why you consider this to be a current or future trend. Support your case with sales data, user interviews, media coverage, industry reports, etc.

03

Identify a market trend

Explain why you consider this to be a current or future trend. Support your case with sales data, user interviews, media coverage, industry reports, etc.

Market trends

1. Explain a current or upcoming market trend

Explain why this is an important trend, how it'll impact your business, and how your competitors are responding to the challenge.

Share the latest sales data or industry metrics that support or signal this trend.

Metric description

00

Metric description

00

Present a case study

Show evidence of the trend. Mention a case study, summarize a report, etc.

LINK TO FULL REPORT



Market trends

1. Identify a current or upcoming market trend

Explain why this is an important trend, how it'll impact your business, and how your competitors are responding to the challenge.

Present a case study

Show evidence of the trend. Mention a case study, summarize a report, etc.

LINK TO FULL REPORT

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Show evidence of the trend. Mention a case study, summarize a report, etc.

LINK TO FULL REPORT

Share the latest sales data or industry metrics that support or signal this trend.

Metric description

00

Metric description

00

Metric description

00

Success factors

Make a claim about what your company needs to do to succeed

Highlight a key attribute of successful brands in your market or industry.

Competitor name

Describe an example of this success factor in action. Explain how it has contributed to the competitor’s business.

Highlight a key attribute of successful brands in your market or industry.

Competitor name

Describe an example of this success factor in action. Explain how it has contributed to the competitor’s business.

Highlight a key attribute of successful brands in your market or industry.

Competitor name

Describe an example of this success factor in action. Explain how it has contributed to the competitor’s business.

Competitive analysis

	YOUR COMPANY	COMPETITOR 1	COMPETITOR 2
PRODUCT OR SERVICE		-	-
PRODUCT OR SERVICE		-	
PRODUCT OR SERVICE		-	
PRODUCT OR SERVICE	-		
PRODUCT OR SERVICE		-	

Success factors

Now you've reviewed the results of the analysis, write a memorable conclusion that sets up your key takeaways.

Summarize a key takeaway. Refer to the concrete findings you presented in the previous slides.

Summarize a key takeaway. Refer to the concrete findings you presented in the previous slides.

Summarize a key takeaway. Refer to the concrete findings you presented in the previous slides.

Key takeaways

Summarize a key takeaway. Reference the findings you presented in the previous slides.

What we found

1. List the data, trends, and factors that support this takeaway
2. Restate your most important findings
3. List any additional details you want your audience to remember

00

Metric description

00

Metric description

A few words about a **relevant trend** that illustrates and supports this takeaway.

Logo

Logo

Main competitors

Next steps

Establish action items based on your key takeaways

Recap a key takeaway	→	Make this takeaway actionable. Suggest what your company or organization can do in response to your analysis.
Recap a key takeaway	→	Make this takeaway actionable. Suggest what your company or organization can do in response to your analysis.
Recap a key takeaway	→	Make this takeaway actionable. Suggest what your company or organization can do in response to your analysis.
Recap a key takeaway	→	Make this takeaway actionable. Suggest what your company or organization can do in response to your analysis.

Any questions? Ask away!

Give your audience space to resolve immediate questions or concerns.